Calculus for Deep Learning

A beginner-friendly guide to derivatives, integrals, chain rule, gradient descent, and backpropagation

Built from intuition first, formulas second.

> **Who this is for.** This resource is for a learner who wants to understand why derivatives matter in deep learning, not just memorize calculus rules. The main goal is to make derivatives, integrals, the chain rule, gradient descent, and backpropagation feel natural.

Contents

The Big Picture: What Calculus Studies

Calculus studies two closely related ideas:

Derivative	How something is changing right now.
Integral	How much has accumulated over an interval.

A derivative answers questions like:

- How fast is the car moving now?
- How steep is this curve at this exact point?
- If I change a neural-network weight slightly, will the loss go up or down?

An integral answers questions like:

- How far did the car travel in total?
- How much water entered the tank?
- How much money was earned after several hours?

The simplest mental model

Think about a water tap filling a bucket.

Derivative thinking	**Integral thinking**
How fast is water flowing right now?	How much water is in the bucket after 10 minutes?
Rate	Total
Liters per minute	Liters

> **Memory sentence.** A derivative is about a **rate**. An integral is about a **total**. If you remember only one thing from this page, remember that.

Position, Speed, and Acceleration

This is the classic example because it makes derivatives feel physical.
Suppose a car's position is recorded over time:

Time (seconds)	Position (meters)	Meaning
0	0	Starting point
1	5	Car has moved 5 meters
2	15	Car has moved 15 meters total
3	30	Car has moved 30 meters total

Now compare positions between nearby times:

- From 0 to 1 second: position changed by 5 meters.
- From 1 to 2 seconds: position changed by 10 meters.
- From 2 to 3 seconds: position changed by 15 meters.

The car is covering more distance each second, so its speed is increasing.

Why position becomes speed

Speed tells us how quickly position changes:

$$\text{Speed} = \frac{\text{change in position}}{\text{change in time}}$$

This is derivative thinking. If position is a function of time, then speed is the derivative of position.

$$\text{Position} \xrightarrow{\text{derivative}} \text{Speed}$$

Why speed becomes acceleration

Acceleration tells us how quickly speed changes:

$$\text{Acceleration} = \frac{\text{change in speed}}{\text{change in time}}$$

So:

$$\text{Position} \xrightarrow{\text{derivative}} \text{Speed} \xrightarrow{\text{derivative}} \text{Acceleration}$$

> This does not mean derivatives are only about cars. It means derivatives always measure **how one quantity changes with respect to another**. In physics, position changing with time gives speed. In deep learning, loss changing with a weight gives a gradient.

What a Derivative Really Means

A derivative measures how much the output of a function changes when the input changes by a tiny amount. Suppose:

$$f(x) = x^2$$

If we increase x, the output x^2 changes. The derivative tells us how sensitive x^2 is to changes in x.

The definition

The derivative of $f(x)$ is defined as:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Read this slowly:

- $x + h$: move a little bit from x.
- $f(x+h) - f(x)$: see how much the output changed.
- Divide by h: compare output change to input change.
- Let $h \to 0$: make the movement tiny, almost zero.

Proof that the derivative of x^2 is $2x$

Let $f(x) = x^2$. Then:

$$f'(x) = \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}$$

Expand:

$$(x+h)^2 = x^2 + 2xh + h^2$$

Substitute:

$$f'(x) = \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$

Cancel x^2:

$$f'(x) = \lim_{h \to 0} \frac{2xh + h^2}{h}$$

Factor h:

$$f'(x) = \lim_{h \to 0} \frac{h(2x+h)}{h}$$

Cancel h:

$$f'(x) = \lim_{h \to 0} (2x + h)$$

Let $h \to 0$:

$$\boxed{f'(x) = 2x}$$

What does $2x$ mean?

If $x = 3$, then:

$$f'(3) = 2(3) = 6$$

This means that near $x = 3$, a tiny increase in x causes the output to increase about 6 times as much.
For example, move from 3 to 3.01:

$$3^2 = 9$$

$$3.01^2 = 9.0601$$

The output changed by 0.0601, while input changed by 0.01:

$$\frac{0.0601}{0.01} = 6.01$$

Very close to 6.

Derivative as Slope: The Hill Analogy

A derivative is the slope of a graph at one point.
Imagine the graph $y = x^2$ as a hill or valley:

Location	Derivative	Meaning
$x = -3$	-6	Slope is negative; moving right goes downhill.
$x = 0$	0	Flat at the bottom.
$x = 3$	6	Slope is positive; moving right goes uphill.

Positive derivative

If the derivative is positive, the graph rises as you move to the right.

$$f'(x) > 0 \quad \Rightarrow \quad \text{moving right increases the output}$$

If you are trying to minimize the output, a positive derivative suggests moving left.

Negative derivative

If the derivative is negative, the graph falls as you move to the right.

$$f'(x) < 0 \quad \Rightarrow \quad \text{moving right decreases the output}$$

If you are trying to minimize the output, a negative derivative suggests moving right.

Zero derivative

If the derivative is zero, the graph is flat at that point.

$$f'(x) = 0$$

This might be a minimum, maximum, or flat saddle-like point. For $y = x^2$, $x = 0$ is the minimum.

> **Deep learning intuition.** A derivative is like a slope sensor. It does not show the entire valley. It tells you the local slope where you currently stand. Gradient descent uses that slope to decide which way to move.

What an Integral Really Means

An integral adds many small pieces together.
If a car travels at a constant speed of 60 km/hour for 5 hours, distance is easy:

$$\text{distance} = 60 \times 5 = 300 \text{ km}$$

But what if speed changes every moment?

Time	Speed
0 min	20 km/hour
10 min	30 km/hour
20 min	40 km/hour
30 min	35 km/hour

Now one multiplication is not enough. We must add the tiny distances traveled over tiny time intervals. That is what an integral does.

$$\text{Total distance} = \int \text{speed}\, dt$$

Salary example

If you earn Rs. 100 per hour, in 30 minutes you earn Rs. 50.
Why? Because 30 minutes is half an hour:

$$100 \times 0.5 = 50$$

This is integral thinking because we are accumulating money over time.

If the rate changes

Suppose your earning rate changes throughout the day. Then total earnings are not just one rate times one time. You must add tiny earnings:

$$\text{total money} = \int \text{earning rate}\, dt$$

> An integral converts a rate into a total. Speed becomes distance. Flow rate becomes water volume. Earning rate becomes total money.

Derivative vs Integral: How to Choose

When you see a problem, ask this question first:

Is the problem asking about change or total?

Choose derivative when the problem asks about change

Use derivatives when the question contains words like:

- rate
- speed
- slope
- sensitivity
- increase/decrease
- optimize
- maximum/minimum
- how fast
- how quickly
- direction of improvement

Examples:

- How fast is the car moving at 4 seconds?
- What is the slope of $y = x^2$ at $x = 3$?
- If I change weight w, how does the loss change?

Choose integral when the problem asks about total

Use integrals when the question contains words like:

- total
- accumulated
- area
- distance from speed
- volume from flow rate
- money from earning rate
- probability over interval

Examples:

- How far did the car travel over 5 hours?
- How much water entered the tank?
- How much money was earned after 8 hours?

Decision table

Question	Nature	Use
How fast is position changing?	Rate	Derivative
How much distance from speed?	Total	Integral
How steep is the graph?	Slope	Derivative
Area under the graph?	Accumulation	Integral
Which weight direction reduces loss?	Optimization	Derivative
Total water from flow rate?	Accumulation	Integral

Golden rule. Derivative: "What is happening right now?" Integral: "What accumulated over time or space?"

The Fundamental Connection

Derivatives and integrals are deeply connected. In many situations, they are inverse operations.

$$\text{Position} \xrightarrow{\text{derivative}} \text{Speed}$$

$$\text{Speed} \xrightarrow{\text{integral}} \text{Position change}$$

This means:

- Differentiate position and you get speed.
- Integrate speed and you get distance traveled.

Example

Suppose:

$$s(t) = t^2$$

where $s(t)$ is position at time t.
Derivative:

$$v(t) = s'(t) = 2t$$

This gives speed.
Now integrate speed from 0 to 4:

$$\int_0^4 2t\,dt = [t^2]_0^4 = 16$$

This gives the distance traveled from $t = 0$ to $t = 4$. It matches:

$$s(4) - s(0) = 16 - 0 = 16$$

The Chain Rule: The Core of Backpropagation

The chain rule is used when one function feeds into another.
Suppose:

$$u = 3x + 1$$
$$y = u^2$$

Then the flow is:

$$x \to u \to y$$

The output y depends on u, and u depends on x. So y depends on x through a chain.

The chain rule formula

$$\boxed{\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}}$$

Read it in English:

The effect of x on y equals the effect of u on y, multiplied by the effect of x on u.

Why multiply?

If Machine A triples something and Machine B doubles it, the total effect is:

$$3 \times 2 = 6$$

Changes pass through stages, so their effects multiply.

Worked example

Let:

$$y = (3x + 1)^2$$

Define:

$$u = 3x + 1$$

Then:

$$y = u^2$$

Now compute:

$$\frac{dy}{du} = 2u$$

$$\frac{du}{dx} = 3$$

Using chain rule:

$$\frac{dy}{dx} = 2u \cdot 3 = 6u$$

Substitute back $u = 3x + 1$:

$$\boxed{\frac{dy}{dx} = 6(3x + 1)}$$

> **Why deep learning cares.** A neural network is a long chain of functions. Layer 1 feeds Layer 2, Layer 2 feeds Layer 3, and so on. Backpropagation uses the chain rule to send derivative information backward through this chain.

Gradient Descent: Learning by Walking Downhill

Gradient descent is the algorithmic version of the hill analogy.

Suppose you are standing on a hill and want to reach the lowest point. You cannot see the whole terrain, but you can feel the slope under your feet.

- If the slope rises to the right, walk left.
- If the slope falls to the right, walk right.
- If the slope is zero, you may be at a flat point.

In math, if we have a loss function $L(w)$, the derivative dL/dw tells us the slope of the loss with respect to weight w. The update rule is:

$$\boxed{w_{new} = w_{old} - \eta \frac{dL}{dw}}$$

where η is the learning rate.

Meaning of the minus sign

The minus sign means we move opposite to the slope.

Derivative	Meaning	What gradient descent does
Positive	Loss rises when w increases	Decrease w
Negative	Loss falls when w increases	Increase w
Zero	Flat slope	Small or no move

Example: minimizing $L(w) = w^2$

The derivative is:

$$\frac{dL}{dw} = 2w$$

Start at:

$$w = 5$$

Let learning rate be:

$$\eta = 0.1$$

Update:

$$w_{new} = 5 - 0.1(2 \cdot 5) = 5 - 1 = 4$$

Next:

$$w_{new} = 4 - 0.1(8) = 3.2$$

Then:

$$3.2 \to 2.56 \to 2.048 \to \cdots$$

The weight moves toward 0, where $L(w) = w^2$ is minimized.

A Tiny Neural Network by Hand

Now we connect derivatives, chain rule, and gradient descent to a tiny neural network.
We want the network to learn this data point:

$$x = 2, \quad y = 10$$

The network has one weight:

$$\hat{y} = wx$$

Start with:

$$w = 3$$

Prediction:

$$\hat{y} = 3 \cdot 2 = 6$$

Actual answer:

$$y = 10$$

Loss:

$$L = (\hat{y} - y)^2 = (6 - 10)^2 = 16$$

Question: should the prediction go up or down?

Try prediction 5:

$$(5 - 10)^2 = 25$$

Loss gets worse.
Try prediction 7:

$$(7 - 10)^2 = 9$$

Loss improves.
So the prediction should increase.

Derivative of loss with respect to prediction

Loss:

$$L = (\hat{y} - y)^2$$

Derivative:

$$\frac{dL}{d\hat{y}} = 2(\hat{y} - y)$$

Plug in values:

$$\frac{dL}{d\hat{y}} = 2(6 - 10) = -8$$

Negative derivative means increasing $\hat{y}$ reduces the loss. This matches our test.

Derivative of prediction with respect to weight

Prediction:

$$\hat{y} = wx$$

Derivative with respect to w:

$$\frac{d\hat{y}}{dw} = x$$

Since $x = 2$:

$$\frac{d\hat{y}}{dw} = 2$$

Increasing the weight increases the prediction.

Chain rule: loss with respect to weight

$$\boxed{\frac{dL}{dw} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dw}}$$

Substitute:

$$\frac{dL}{dw} = (-8)(2) = -16$$

The derivative is negative, so gradient descent will increase w:

$$w_{new} = w_{old} - \eta \frac{dL}{dw}$$

If $\eta = 0.1$:

$$w_{new} = 3 - 0.1(-16) = 3 + 1.6 = 4.6$$

New prediction:

$$\hat{y} = 4.6 \cdot 2 = 9.2$$

New loss:

$$(9.2 - 10)^2 = 0.64$$

The loss dropped from 16 to 0.64. This is learning.

Backpropagation in Plain English

Backpropagation means sending derivative information backward through a chain of functions.
Forward pass:

$$\text{Input} \to \text{Weight} \to \text{Prediction} \to \text{Loss}$$

Backward pass:

$$\text{Loss} \to \text{Prediction derivative} \to \text{Weight derivative}$$

Backpropagation answers:

How much did each parameter contribute to the final loss?

Why the chain rule is enough

A neural network is a chain:

$$x \to z_1 \to a_1 \to z_2 \to a_2 \to \hat{y} \to L$$

Each arrow is a function. To understand how an early weight affects the final loss, we multiply derivatives across the chain.

A simplified version:

$$\frac{dL}{dw} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz} \cdot \frac{dz}{dw}$$

This is just the chain rule repeated.

Deep learning translation

Calculus idea	Deep learning meaning
Function	Layer or model
Derivative	Sensitivity
Loss	Error score
Gradient	Derivative of loss with respect to parameters
Gradient descent	Update parameters to reduce loss
Chain rule	How error signal flows backward through layers
Backpropagation	Efficient chain rule for neural networks

> Backpropagation is not a mysterious AI trick. It is the chain rule applied carefully and efficiently through many layers.

Common Misunderstandings

Misunderstanding 1: A derivative tells you the final answer

No. A derivative tells you the local slope or sensitivity. It says what happens near your current point.

Misunderstanding 2: A negative derivative is bad

No. A negative derivative simply means the function decreases as the input increases. In optimization, that may be very useful.

Misunderstanding 3: Integral means multiplication

Not exactly. Multiplication is a simple integral when the rate is constant. If the rate changes continuously, integration adds many tiny changing pieces.

Misunderstanding 4: Backpropagation is different from calculus

Backpropagation is mostly calculus. More specifically, it is repeated chain rule plus efficient bookkeeping.

Worked Practice Problems

Problem 1: derivative from the power rule

Find the derivative of:

$$f(x) = x^3$$

Using the power rule:

$$\frac{d}{dx}x^n = nx^{n-1}$$

So:

$$f'(x) = 3x^2$$

At $x = 2$:

$$f'(2) = 3(2)^2 = 12$$

Meaning: near $x = 2$, the graph is changing at about 12 output units per 1 input unit.

Problem 2: choosing derivative or integral

A tap pours water at 4 liters per minute for 10 minutes. How much water entered the bucket?
This asks for total water, so use integral thinking:

$$4 \times 10 = 40 \text{ liters}$$

Problem 3: chain rule

Find the derivative of:

$$y = (5x - 2)^2$$

Let:

$$u = 5x - 2$$

Then:

$$y = u^2$$

Compute:

$$\frac{dy}{du} = 2u$$

$$\frac{du}{dx} = 5$$

Chain rule:

$$\frac{dy}{dx} = 2u \cdot 5 = 10u$$

Substitute back:

$$\boxed{\frac{dy}{dx} = 10(5x - 2)}$$

Problem 4: one weight neural network

Let:

$$\hat{y} = wx, \quad L = (\hat{y} - y)^2$$

Given:

$$x = 3, \quad y = 12, \quad w = 2$$

Prediction:

$$\hat{y} = 2 \cdot 3 = 6$$

Loss:

$$L = (6 - 12)^2 = 36$$

Derivative of loss with respect to prediction:

$$\frac{dL}{d\hat{y}} = 2(6 - 12) = -12$$

Derivative of prediction with respect to weight:

$$\frac{d\hat{y}}{dw} = x = 3$$

Chain rule:

$$\frac{dL}{dw} = (-12)(3) = -36$$

Since derivative is negative, gradient descent increases w.

Practice Exercises

Try these without looking at the answers first.

A. Concept questions

1. If a question asks "how fast is this changing?", should you think derivative or integral?
2. If a question asks "how much accumulated?", should you think derivative or integral?
3. If a derivative is positive and you want to minimize the function, should you move left or right?
4. If a derivative is negative and you want to minimize the function, should you move left or right?
5. Why does the chain rule multiply derivatives instead of adding them?

B. Calculation questions

1. Find the derivative of x^4.
2. Find the derivative of $(2x+1)^2$.
3. A car travels at 50 km/hour for 3 hours. How far does it travel?
4. A model predicts $\hat{y} = 4$, actual $y = 9$, and $L = (\hat{y} - y)^2$. Should prediction increase or decrease?
5. For $\hat{y} = wx$, if $x = 2$, $w = 1$, and $y = 8$, compute dL/dw for $L = (\hat{y} - y)^2$.

Answers

A. Concept answers

1. Derivative.
2. Integral.
3. Move left, because positive derivative means moving right increases the function.
4. Move right, because negative derivative means moving right decreases the function.
5. Because each stage scales the change from the previous stage. Effects through a chain multiply.

B. Calculation answers

1. $4x^3$.
2. Let $u = 2x + 1$. Then $dy/dx = 2u \cdot 2 = 4(2x + 1)$.
3. $50 \times 3 = 150$ km.
4. Prediction should increase. 4 is below 9, and increasing prediction reduces squared error.
5. Prediction: $\hat{y} = 1 \cdot 2 = 2$. Loss derivative: $dL/d\hat{y} = 2(2 - 8) = -12$. Prediction derivative: $d\hat{y}/dw = 2$. So $dL/dw = -24$. Gradient descent increases w.

Final Cheat Sheet

Core meanings

Concept	Meaning
Function	A machine that converts input into output.
Derivative	How output changes when input changes slightly.
Integral	Total accumulated amount from many tiny pieces.
Chain rule	How changes multiply through connected functions.
Gradient	Derivative of loss with respect to parameters.
Gradient descent	Move parameters opposite the gradient to reduce loss.
Backpropagation	Chain rule applied backward through a neural network.

Important formulas

$$\frac{d}{dx}x^n = nx^{n-1}$$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$w_{new} = w_{old} - \eta \frac{dL}{dw}$$

$$L = (\hat{y} - y)^2$$

$$\frac{dL}{d\hat{y}} = 2(\hat{y} - y)$$

Decision guide

Question type	Think
How fast?	Derivative
How steep?	Derivative
How sensitive?	Derivative
Which direction improves loss?	Derivative/gradient
How much total?	Integral
How much accumulated?	Integral
Area under curve?	Integral
Distance from speed?	Integral

Deep learning summary

A neural network learns by repeatedly doing this:

1. Make a prediction.
2. Compute the loss.
3. Use derivatives to measure how each weight affects the loss.
4. Use chain rule to connect effects through layers.
5. Update weights in the direction that reduces loss.
6. Repeat many times.

> **Final intuition.** Derivatives are the steering mechanism of deep learning. They tell the model which direction to adjust its weights. The chain rule lets that steering signal travel through many layers. Gradient descent uses that signal to reduce loss step by step.